

SynthID Forensic Analysis Report

A (Yours): your_output.jpg | B (Theirs): working_tool_output.jpg

1. File & Metadata

Property	Yours (A)	Theirs (B)
File size	831.84 KB	398.83 KB
Dimensions	1376x768	1600x899
Format	JPEG	JPEG
EXIF tags present	0	0

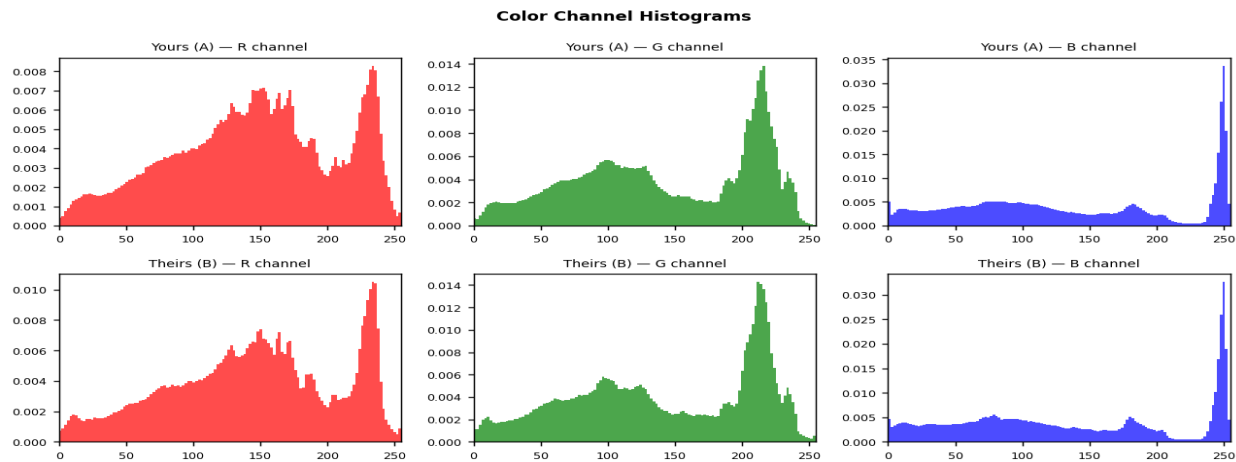
■ File size differs by 433.0 KB. Larger file often means less aggressive JPEG compression was applied — which preserves more of the original pixel structure including the watermark.

2. Pixel-Level Difference Statistics

Metric	Value	Meaning
Mean absolute diff	10.3615 / 255	Average per-pixel change. Target: 3–8
Max diff	234.0 / 255	Largest single pixel change
Std of diff	17.5123	Spread of changes across image
PSNR	23.26 dB	Quality loss. >35dB = invisible, <25dB = visible
R channel mean diff	0.4103	
G channel mean diff	0.2871	
B channel mean diff	0.3563	

3. Color Channel Histograms

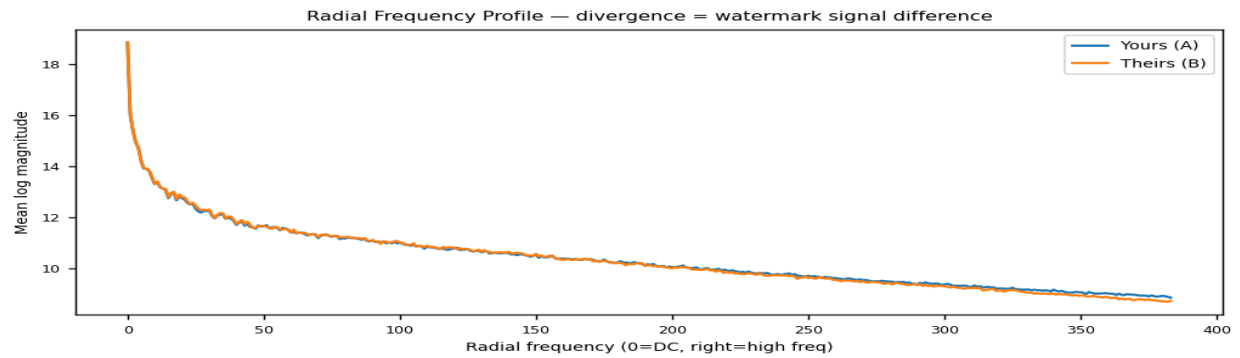
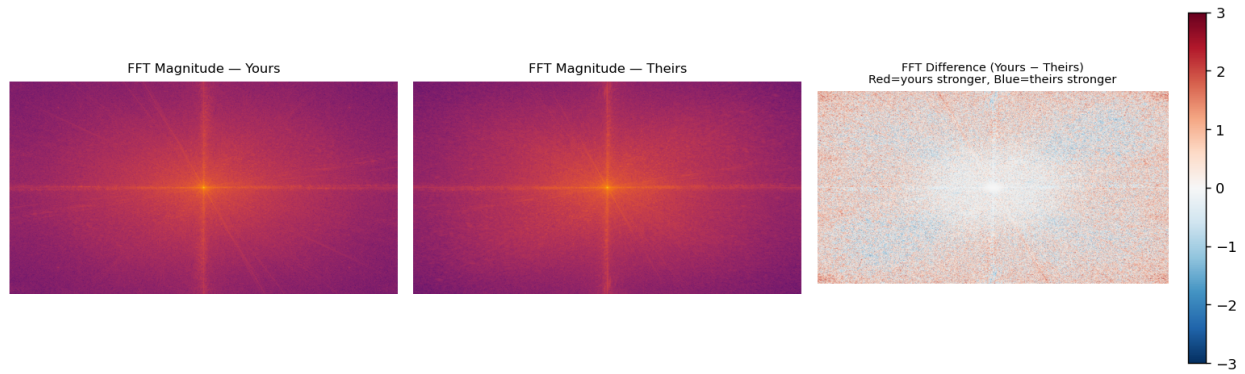
Differences in histogram shape between A and B indicate colour processing differences. SynthID removal tools often subtly shift channel distributions as a side effect of their perturbations.



4. Frequency Domain (FFT) Analysis

This is the most important section. SynthID is embedded in the mid-frequency bands of the image. The FFT magnitude maps show where each image has energy. The difference map shows which tool disrupted more frequency content — and where. The radial profile shows the exact frequency bands each tool is targeting.

Frequency Domain (FFT) Analysis — SynthID lives here

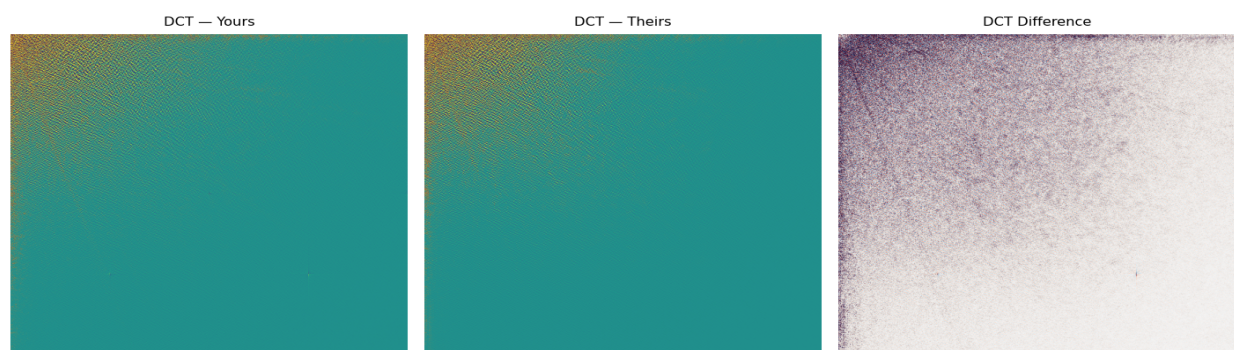


FFT Metric	Yours (A)	Theirs (B)
Mean magnitude	8.9875	8.8456
Std magnitude	1.1439	1.2367
Mid-freq diff (A-B)	0.6659	↑ higher = more disruption
Overall FFT diff	0.6582	↑ higher = more disruption

5. DCT Coefficient Analysis

JPEG and SynthID both operate in the DCT domain. SynthID modifies mid-range DCT coefficients. If the working tool is disrupting different DCT bands than yours, that gap is likely why their removal works and yours doesn't.

DCT Coefficient Analysis

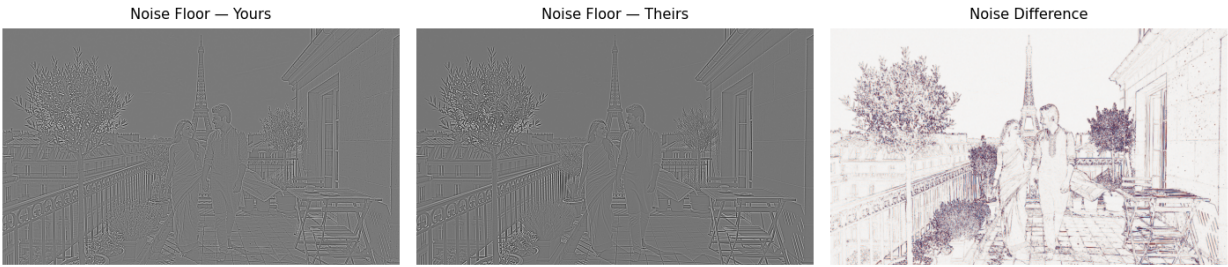


DCT Band	Diff magnitude	Notes
Low freq (DC area)	46.516327	Should be LOW — don't touch DC
Mid freq (target)	16.032919	Should be HIGH — this is SynthID's home
High freq	4.263699	Less important for SynthID
Overall mean	10.823529	

6. Noise Floor Analysis

The noise floor shows what each tool leaves behind after processing. SynthID tools typically add a calibrated noise layer — too little and the watermark survives, too much and the image looks grainy.

Noise Floor Analysis



Noise Metric	Yours (A)	Theirs (B)
Noise std dev	11.5034	10.9073
Mean abs noise	6.4235	5.7160
Noise diff std	10.5472	↑ = different noise patterns

7. Texture & Entropy Analysis

Metric	Yours (A)	Theirs (B)	Notes
Entropy	7.7157	7.7281	Higher = more randomness
Contrast	0.4116	0.3793	Local intensity variation
Homogeneity	0.8724	0.8836	Smoothness of texture
Energy	0.3436	0.3532	Uniformity of texture
Correlation	0.9352	0.9415	Linear dependency of grey levels

8. Visual Difference Heatmap

Bright areas in the heatmap show where the two images differ most. A good SynthID removal tool produces changes spread evenly across the entire image — not concentrated in corners or specific regions.

Visual Difference Map



9. Key Findings & Recommendations

- **PIXEL DIFF:** Pixel difference looks reasonable (10.36/255).
- **FFT MID-FREQ:** Good mid-frequency disruption (0.6659). This is the target range for SynthID removal.
- **DCT COEFFS:** Low-freq DCT change (46.5163) > mid-freq (16.0329). You are changing the wrong coefficients — SynthID is in mid-range DCT, not low-freq.
- **NOISE FLOOR:** Noise floors are similar (yours 11.503, theirs 10.907).

■ ENTROPY: Entropy similar (yours 7.716, theirs 7.728).

What to change in your script:

1. Your DCT perturbation is hitting the wrong bands. Target DCT indices $[h//8:h//2, w//8:w//2]$ — avoid the DC and very-low coefficients.